Riesz Representation in Hilbert Spaces

An End-to-End Meta-Math Walkthrough

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(Dated: October 2026)

Abstract

The kernel has an orthogonal direction that captures every value of the functional; scaling that direction gives the representing vector. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

FA-03 / Functional Analysis and Measure Theory / Report v1

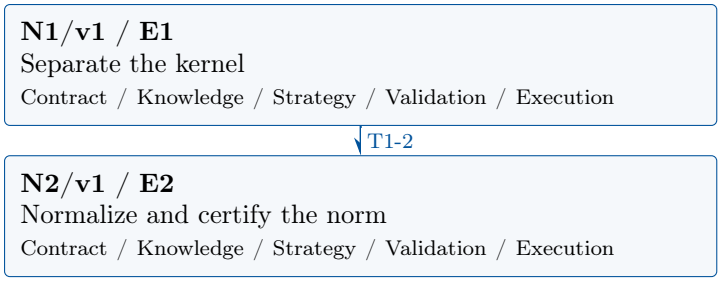
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For a real or complex Hilbert space H with inner product linear in its first variable, every bounded linear functional $\ell : H \to \mathbb{F}$ has a unique $y \in H$ such that $\ell(x) = \langle x, y\rangle$ for all $x \in H$. Also $\|\ell\| = \|y\|$.*

A. Foundations and permitted imports

Import FA-02: every vector has a unique orthogonal decomposition relative to a closed linear subspace. Use Cauchy–Schwarz and continuity of bounded linear maps. The operator norm is $\|\ell\| = \sup_{\|x\|\le 1} |\ell(x)|$. This is the Hilbert-space Riesz theorem, not the measure representation theorem on spaces of continuous functions.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. There is $z \in K^{\perp}$ with $\ell(z) \neq 0$, and $x - \ell(x)z/\ell(z) \in K$ for every x.

Receiving activity. N2.1

Compatibility. E1/v1; the first-variable linear convention. Check: The scalar in the second inner-product argument is conjugated; omitting it changes the represented functional over $\mathbb{C}$.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Separate the kernel

Object and role. Project a vector outside the kernel and retain its nonzero perpendicular remainder.
Local obligation: There is $z \in K^\perp$ with $\ell(z) \neq 0$, and $x - \ell(x)z/\ell(z) \in K$ for every x.

Inputs. A nonzero bounded functional ℓ; its kernel K.

C — Contract

Publish E1/v1 only after establishing this obligation: There is $z \in K^\perp$ with $\ell(z) \neq 0$, and $x - \ell(x)z/\ell(z) \in K$ for every x.

K — Knowledge

A nonzero bounded functional ℓ; its kernel K. Retain the stated assumptions and import versions.

S — Strategy

Project a vector outside the kernel and retain its nonzero perpendicular remainder.

V — Validation

The remainder cannot vanish or lie in the kernel, because its functional value equals the original nonzero value.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 If $\ell = 0$, take $y = 0$ and the representation and norm equality are immediate; uniqueness follows from testing the difference against itself as in N2. For the remaining case choose w with $\ell(w) \neq 0$. Continuity makes K closed, so FA-02 gives $w = P_K w + z$ with $z \perp K$.

N1.2 Linearity gives $\ell(z) = \ell(w) \neq 0$. For each x, set $k = x - \ell(x)z/\ell(z)$. Then $\ell(k) = 0$, so $k \in K$. Thus $\langle x, z\rangle = \ell(x)\|z\|^2/\ell(z)$.

Export E1/v1. There is $z \in K^\perp$ with $\ell(z) \neq 0$, and $x - \ell(x)z/\ell(z) \in K$ for every x.

Acceptance gate. The remainder cannot vanish or lie in the kernel, because its functional value equals the original nonzero value.

Targeted repair. If division by $\ell(z)$ occurs before nonvanishing is shown, restore the separate zero-functional case and the equality $\ell(z) = \ell(w)$. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Normalize and certify the norm

Object and role. Choose the conjugate scalar in the second argument, then test at the normalized representing vector. **Local obligation:** A unique y represents ℓ, and $\|\ell\| = \|y\|$.

Inputs. E1/v1; the first-variable linear convention. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: A unique y represents ℓ, and $\|\ell\| = \|y\|$.

K — Knowledge

E1/v1; the first-variable linear convention. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Choose the conjugate scalar in the second argument, then test at the normalized representing vector.

V — Validation

The scalar in the second inner-product argument is conjugated; omitting it changes the represented functional over $\mathbb{C}$.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 Define $y = \overline{\ell(z)}z/\|z\|^2$ (drop conjugation over $\mathbb{R}$). Then $\langle x, y\rangle = \ell(z)\langle x, z\rangle/\|z\|^2 = \ell(x)$ by E1.

N2.2 If y_1, y_2 represent the same functional, $\langle x, y_1 - y_2\rangle = 0$ for all x. Taking $x = y_1 - y_2$ proves equality. Cauchy–Schwarz gives $\|\ell\| \le \|y\|$; if $y \neq 0$, testing $x = y/\|y\|$ gives the reverse inequality. The zero case has both norms zero.

Export E2/v1. A unique y represents ℓ, and $\|\ell\| = \|y\|$.

Acceptance gate. The scalar in the second inner-product argument is conjugated; omitting it changes the represented functional over $\mathbb{C}$.

Targeted repair. If conjugation is absent, recompute the second-argument scalar and check the one-dimensional complex example before releasing the norm and representation. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Separate the kernel

1. **N1.1.** If $\ell = 0$, take $y = 0$ and the representation and norm equality are immediate; uniqueness follows from testing the difference against itself as in N2. For the remaining case choose w with $\ell(w) \neq 0$. Continuity makes K closed, so FA-02 gives $w = P_K w + z$ with $z \perp K$.

2. **N1.2.** Linearity gives $\ell(z) = \ell(w) \neq 0$. For each x, set $k = x - \ell(x)z/\ell(z)$. Then $\ell(k) = 0$, so $k \in K$. Thus $\langle x, z\rangle = \ell(x)\|z\|^2/\ell(z)$.

B. N2 — Normalize and certify the norm

1. **N2.1.** Define $y = \overline{\ell(z)}z/\|z\|^2$ (drop conjugation over $\mathbb{R}$). Then $\langle x, y\rangle = \ell(z)\langle x, z\rangle/\|z\|^2 = \ell(x)$ by E1.

2. **N2.2.** If y_1, y_2 represent the same functional, $\langle x, y_1 - y_2\rangle = 0$ for all x. Taking $x = y_1 - y_2$ proves equality. Cauchy–Schwarz gives $\|\ell\| \leq \|y\|$; if $y \neq 0$, testing $x = y/\|y\|$ gives the reverse inequality. The zero case has both norms zero.

□

V. WORKED ORCHESTRATION

On $\mathbb{C}^2$ with $\langle x, y\rangle = x_1\overline{y_1} + x_2\overline{y_2}$, let $\ell(x) = x_1 + 2ix_2$. Its representing vector is $y = (1, -2i)$, so $\|\ell\| = \sqrt{5}$. The unit vector $y/\sqrt{5}$ attains the operator norm. This checks the conjugation convention directly.

A. Boundary and hypothesis audit

Completeness matters: on c_{00} with the ℓ^2 norm, $\ell(x) = \sum_n x_n/n$ is bounded by Cauchy–Schwarz. A representing vector within c_{00} would have coordinates $1/n$ by testing the standard basis, an impossibility. The completion ℓ^2 supplies that vector.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N2 candidate: $y = \ell(z)z/\|z\|^2$ under the first-variable linear convention. For $H = \mathbb{C}$, $\ell(x) = ix$, and $z = 1$, it gives $\langle x, i\rangle = -ix$. **Diagnosis:** Knowledge used the wrong scalar convention. **Repair:** conjugate $\ell(z)$ as in N2.1. The statement is unchanged; recheck both the representation and its norm certificate.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Apply the representation to bounded variational functionals, as in the weak Poisson existence project.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. MIT OpenCourseWare, Introduction to Functional Analysis: lecture notes. See the lectures on integration, Hilbert spaces, Baire category, and Hahn–Banach.